

Akash S Kumbar

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OVERVIEW

I build computer vision and robotics systems from the math up, deriving the algorithms, designing and training the networks, and making them work on physical robots. In my IROS 2026 paper, Beyond Frontiers, I reframed autonomous exploration as scene-anomaly minimization, enabling a robot to seek out poorly reconstructed regions while mapping unknown real-world environments. I also built an end-to-end navigation stack for a Husky A200, fusing RGB-D and LiDAR perception with localization, global planning, and MPC-based reactive avoidance; it prevented collisions in 95% of more than 20 dynamic-obstacle trials. My research also spans dynamic urban-scene reconstruction with Gaussian splatting and SDFs at ICCV 2025, and two first-author ICCV workshop papers on point-cloud upsampling. Across these projects, I have focused on carrying ideas through the full journey from equations and experiments to code and hardware.

EDUCATION

- IIIT-Hyderabad** Hyderabad, India
• *MS(Research), Computer Science Engineering; CGPA: 9.33/10.0* 2024 – present
- KLE Technological University** Hubballi, India
• *B.E., Electronics and Communication Engineering; CGPA: 8.42/10.0* 2019 – Jul 2023

PUBLICATIONS

* indicates equal contribution.

- **Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration** ↗ : IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
Authors: Akash Kumbar, Abhinav Raundhal, Madhava Krishna.
We introduce SCAGE, an autonomous exploration framework that identifies geometric inconsistencies in 3D point clouds to guide robotic navigation toward poorly reconstructed regions, treating exploration as anomaly minimization rather than frontier-chasing for superior coverage and reconstruction fidelity.
- **Leveraging 2D Priors and SDF Guidance for Dynamic Urban Scene Rendering** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
Authors: Siddharth Tourani, Jayaram Reddy, **Akash Kumbar**, Satyajit Tourani, Nishant Goyal, Madhava Krishna, N Dinesh Reddy, Muhammad Haris Khan.
We propose a unified framework that integrates Signed Distance Functions with 3D Gaussian Splatting, enabling state-of-the-art dynamic scene reconstruction and rendering without reliance on LiDAR or ground-truth motion annotations.
- **ASUR3D: Arbitrary Scale Upsampling and Refinement of 3D Point Clouds Using Local Occupancy Fields** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
Authors: Akash Kumbar, Tejas Anvekar, Ramesh Ashok Tabib, Uma Mudenagudi.
A novel framework for arbitrary-scale upsampling of 3D point clouds. Leveraged Local Occupancy Fields to learn continuous implicit representations, enabling the reconstruction of intricate geometric details from sparse, noisy inputs.
- **TP-NoDe: Topology-Aware Progressive Noising and Denoising of Point Clouds Towards Upsampling** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
*Authors: Akash Kumbar**, Tejas Anvekar*, Tulasi Amitha Vikrama, Ramesh Ashok Tabib, Uma Mudenagudi.
Proposed TP-NoDe, a topology-aware Progressive Noising and Denoising technique for 3D point cloud upsampling. For topology awareness, we proposed Density Aware k-Nearest Neighbours (DA-kNN).

EXPERIENCE

- **Robotics Research Center, IIIT-Hyderabad** Hyderabad, India
Graduate Researcher Jul 2023 – present
 - **Dynamic scene reconstruction:** Researching on dynamic scene reconstruction, 3D scene representations, and anomaly detection in LiDAR point clouds.
 - **Publications:** Published a paper in ICCV 2025 and a paper accepted to IROS 2026.
- **Center of Excellence in Visual Intelligence (CEVI), KLE Technological University** Hubballi, India
Student Researcher; Research Intern (Jan – May 2023) Aug 2021 – May 2023
 - **Point cloud refinement:** Conducted research on point-cloud refinement under Prof. Uma Mudenagudi, developing neural methods for arbitrary-scale upsampling and topology-aware refinement.
 - **Publications:** This work resulted in two first-author papers at the ICCV 2023 Workshops: ASUR3D and TP-NoDe.

PROJECTS

- **Collision Avoidance in Unknown Environments via Multimodal Perception and Reactive Planning** ↗ : Developed a real-time navigation framework for the Husky UGV that fuses LiDAR and RGB-D data to autonomously detect and track dynamic obstacles in unknown environments without prior maps. By integrating SLAM-based localization with a reactive planner based on Model Predictive Control, successfully implemented collision avoidance against moving agents during real-world deployments.
Tech Stack: ROS, PyTorch, RGB-D, LiDAR, Fast-LIO, SLAM, MPC, Gazebo
- **ARROU: Adversarially Robust Rank-One Unlearning for LLMs** ↗ : Developed an attacker-in-the-loop framework that combines latent adversarial training with rank-one model editing to reliably remove harmful knowledge from LLMs.
Tech Stack: Python, PyTorch, Hugging Face Transformers, Llama 3.2-1B, CUDA
- **MAST3R-SLAM** ↗ : Developed a SLAM pipeline from scratch as an experimental framework to evaluate how MAST3R features perform in a SLAM setting. Implemented core components including feature extraction, tracking, and mapping to analyze their effectiveness in reconstruction and localization.
Tech Stack: PyTorch, CUDA
- **Camera-Based Vehicle Functions (Bosch PRIXEL)** ↗ : Designed a computer vision solution to detect potholes and speed breakers on roads, utilizing a custom-engineered dataset that was collected and annotated in-house.
Tech Stack: Python, PyTorch, ONNX, OpenCV
- **Learning-Based Refinement of 3D point clouds through hole filling** ↗ : Developed a deep learning model to identify and repair point cloud gaps caused by occlusion and reflections. Designed an algorithm to generate datasets with realistic holes in point clouds for training and testing.
Tech Stack: Python, PyTorch, PyTorch3D, Open3D

SKILLS SUMMARY

- **Programming:** Python, C++, C, CUDA, Embedded C
- **Machine Learning & Deployment:** PyTorch, TensorFlow, ONNX, scikit-learn, Hugging Face Transformers, Model Training, Evaluation Pipelines, Real-Time Inference
- **Computer Vision & 3D Perception:** Object Detection, Instance Segmentation, Tracking, 3D Scene Understanding, 3D Geometry, Point-Cloud Processing, Scene Reconstruction, Gaussian Splatting, SDFs
- **Robotics & Multimodal Sensor Fusion:** ROS, SLAM, Fast-LIO, LiDAR, RGB-D, Path Planning (OMPL, RRT*), MPC
- **Data:** Data Processing, Dataset Curation and Management, Annotation, Synthetic Data Generation, Data Augmentation
- **Simulation:** Gazebo, CARLA, Habitat
- **Robotics Hardware & Sensors:** Husky A200, Intel RealSense, Ouster OS1, Livox Mid-70

TEACHING AND SERVICE

- **Teaching Assistant:** Mobile Robotics, IIIT-Hyderabad — Monsoon 2026
- **Instructor:** RRC Summer School, IIIT-Hyderabad (2025 – 2026) — Taught deep learning for 3D vision, neural representations, and 3D Gaussian Splatting; 2026 lecture recording ↗
- **Reviewer:** IROS 2026 Workshop

HONORS AND AWARDS

- Certificate of appreciation in Bosch's PRIXEL Program, Bosch Global Software Technologies — Awarded for developing a robust, real-time computer vision-based speed breaker and pothole detection model.